

Sai Anudeep Kashetty

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PROFESSIONAL SUMMARY

- **Data Scientist with over 5 years of experience** delivering end-to-end machine learning solutions across finance, healthcare, and technology, improving churn, fraud detection, and customer retention.
- Specialized in **supervised/unsupervised learning**, deep learning with **PyTorch** and **TensorFlow**, time series forecasting, and optimizing models for scalability and performance.
- Deployed ML models on **AWS** and **Azure** using MLOps best practices, including automated **CI/CD pipelines**, model performance monitoring, and stability validation.
- Experienced in financial and healthcare domains, including fraud detection, customer retention, and digital payment systems. Skilled in experimentation (**A/B testing**), **GCP (BigQuery)**, and stakeholder-facing analytics using Tableau
- Translated complex analytics into strategic insights through **Tableau**, **Power BI**, and interactive dashboards
- **Committed to responsible AI**, ensuring model transparency and regulatory compliance (AML, audit) using SHAP, LIME, and robust validation techniques.
- **Building solutions with Generative AI and LLMs** (GPT-4, Gemini, FAISS, RAG) for recommendation engines and semantic search applications.

WORK EXPERIENCE

Citizens | Data Scientist | Plano, TX

Jan 2024 – Present

- Developed and deployed **predictive models** for financial churn risk and revenue forecasting using **Artificial Neural Networks (PyTorch)**, achieving **88% AUC** and driving a **15%** increase in campaign ROI.
- Optimized **supervised ML algorithms** such as **Random Forest**, **XGBoost**, and **SVM**, as well as **unsupervised models** like **K-Means** and **Hierarchical Clustering** for **customer segmentation**, **fraud detection**, and **churn prediction**.
- Architected **Natural Language Processing (NLP)** pipelines using **Word2Vec**, **GloVe**, **Scikit-learn**, and **Keras** to extract insights from over 500K customer feedback records, enhancing customer experience analytics.
- Executed extensive **feature engineering**, including **feature selection**, **encoding**, **scaling**, and **outlier detection** with **Pandas** and **NumPy** to build clean, scalable datasets supporting ML training.
- Automated **ETL pipelines** using **Python** and **MySQL**, enabling **near real-time data processing** to feed ML models and analytics.
- Built and deployed fraud detection and financial capacity forecasting models on **AWS EC2**, enabling real-time scoring with **12%** fewer false positives across digital payment and audit pipelines.
- Applied **MLOps** best practices (CI/CD, drift detection, performance monitoring) to maintain robust production models in finance and compliance systems.
- Ensured **model interpretability** and regulatory compliance using **SHAP** and **LIME** for explainability in high-stakes environments.
- Worked in **cross-functional**, multicultural teams to develop innovative ML solutions with an emphasis on ethical AI and fairness.
- Managed the **full ML lifecycle**: data acquisition, exploratory data analysis (EDA), model development, hyperparameter tuning, deployment, and post-deployment monitoring.

Optum | Data Scientist | Hyderabad, India

Sep 2021 – Oct 2022

- Developed supervised **machine learning models** for health plan marketing using claims and EHR data, optimizing performance via **GridSearchCV**, hyperparameter tuning, and feature selection, achieving up to **85% accuracy**.
- Built and deployed gradient boosting models (**XGBoost**) for patient readmission prediction with explainability via SHAP, improving intervention targeting accuracy.
- Implemented NLP models for entity recognition in clinical notes and integrated predictions into Azure ML dashboards for provider decision support.
- Performed exploratory data analysis (EDA) on large-scale clinical datasets with **PySpark** and **Pandas**, enabling robust feature engineering and improving patient stratification by **20%**.

- Integrated structured and semi-structured data from **SAS, SQL, Hive**, and data lakes, streamlining data pipelines and reducing processing time by **25%**.
- Designed customized Tableau visualizations to highlight care gap patterns and boost data-driven strategy in patient engagement initiatives.
- Performed **A/B testing** on marketing and care outreach campaigns, generating insights that improved patient engagement and intervention effectiveness.
- Evaluated model performance using **ROC curves, AUC**, and **K-fold cross-validation** to ensure robustness, generalizability, and interpretability in clinical settings.
- Deployed predictive models on **Azure**, enabling scalable, real-time scoring and seamless integration into healthcare analytics platforms.

Trident Soft Solutions | Data Analyst | Hyderabad, India

May 2020 – Aug 2021

- Developed and optimized ETL pipelines using **SSIS**, integrating multi-source datasets and improving data processing efficiency by **20%**.
- Extracted, cleaned, and transformed large datasets using **Python (Pandas, NumPy)**, automating processes to enhance data reliability and workflow efficiency.
- Structured and maintained **relational database schemas**, supporting scalable analytics and data integration projects.
- Wrote complex **SQL queries** to generate actionable business reports, reducing report generation time and supporting strategic decision-making.
- Created interactive **Tableau dashboards** and **Excel pivot tables**, driving **18%** increase in stakeholder adoption of business intelligence tools.
- Led in-depth **data validation** and cleansing to ensure accuracy, consistency, and compliance across reporting systems.
- Performed **trend and root cause analysis** on operational data to identify process improvements and uncover cost-saving opportunities.
- Delivered **data-driven presentations** to cross-functional teams and stakeholders, influencing key business decisions and project prioritization.

ACADEMIC PROJECTS

Social Distance Detection Using Deep Learning

- Engineered a **TensorFlow + OpenCV** model for real-time social distance monitoring with **YOLO V3** Object detection architecture and achieved high model accuracy in real-world scenarios.

AI-Driven Product Recommendation System

- Constructed a personalized shopping recommendation engine using **Sentence Transformers, FAISS** for **vector search**, **GPT-4** to generate product suggestions, and integrated LLMs to deliver user-specific recommendations.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning, Time Series Forecasting, Generative AI

Frameworks & Libraries: TensorFlow, OpenCV, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, PySpark

Generative AI and LLMs: RAG, OpenAI API (GPT-3.5 Turbo, GPT-4.0), Gemini 1.5, FAISS, Pinecone, ChromaDB

Big Data: Apache Spark, Hadoop, MapReduce, Hive, HDFS, PySpark

Cloud: AWS (SageMaker, EC2, S3, Athena, Glue), Azure (Data Factory, Blob Storage), GCP (BigQuery, Vertex AI).

Databases: SQL Server, MySQL, PostgreSQL, NoSQL, Oracle

Data Visualization: Tableau, Power BI, Excel, Matplotlib, Seaborn

Statistical Analysis: Hypothesis Testing, T-Test, Descriptive Statistics, Correlation Techniques

NLP: NLTK, SpaCy, BERT, Transformers, Hugging Face, NER, Text Classification

EDUCATION

Master of Science, Data Science – GPA: 3.6/4

University of North Texas, Denton, Texas

Jan 2023 - Dec 2024